

Appendix A

Development of and Justification for Current Assumptions

This appendix describes the development of and justification for the current actuarial assumptions used in this filing.

The persistency and morbidity assumptions were developed using historical experience and predictive analytics. Where actual experience was limited or did not exist, industry experience and actuarial judgment was also used. The experience used to develop these assumptions includes historical experience of the policies originally issued by Combined Insurance Company of America and Combined Life Insurance Company of New York. It also includes the experience of other business issued by MedAmerica Insurance Company (MAPA); MAPA's two sister companies, MedAmerica Insurance Company of Florida (MAFL) and MedAmerica Insurance Company of New York (MANY) (MAPA, MAFL, and MANY collectively referred to as MedAmerica); and MedAmerica's affinity partners that issued the same products. Additionally, experience on other blocks of business originally issued by MedAmerica and its affinity partners (including commuted business), and MedAmerica's other acquired blocks of business is also used.

The persistency and morbidity assumptions were developed on a first principles basis. In this context, "first principles" means developing key assumptions (namely, morbidity and mortality) at the component level and modeling active and disabled lives separately. Separate assumptions were developed for: (1) claim incidence, (2) voluntary lapse, (3) active mortality, (4) claim termination (including separate assumptions for disabled mortality and recovery), and (5) utilization. For each assumption except utilization, experience from 2008 through 2017 with runout through 2018 was used. The utilization assumption was developed based on historical experience from 2016 through 2019 with runout through March 31, 2020 to capture more recent information as cost of care and utilization trends can fluctuate over short periods of time. Experience adjustment factors were developed using predictive analytics as described in the Predictive Analytics section below. The assumptions outlined in this Appendix were reviewed with experience from 2010 through 2019; no changes were made as a result of this review.

As an additional consideration, the ongoing COVID-19 pandemic has created uncertainty regarding future long-term care (LTC) experience. The short-term impact on LTC assumptions may be favorable due to higher mortality (particularly at advanced ages) and potentially lower incidence and benefit utilization. However, due to the long duration nature of LTC insurance, minor deviations in experience over a relatively small number of calendar years are not expected to have a significant impact on lifetime projections. Therefore, we did not make any adjustments to these assumptions to capture anticipated impacts of the pandemic. However, we will continue to monitor emerging experience and trends so that adjustments can be included in the future if believed to be representative of long-term trends.

Improvement assumptions for active mortality and claim incidence were developed based on individual annuitant or industry experience, along with actuarial judgment. The rate increase dependent assumptions were developed using historical experience, and actuarial judgment where experience was limited or did not exist.

The sections that follow provide more detail on the development of and justification for the current assumptions that are material to the projections in this filing.

Active Assumptions

The assumptions for active (i.e., healthy) lives were developed based on detailed historical experience from 2008 through 2017 with runout through 2018 for MedAmerica's organic (including affinity partners) and acquired business. Experience adjustment factors were developed using predictive analytics as described in the Predictive Analytics section below. The active assumptions include (1) claim incidence (the probability that an active life becomes disabled), (2) voluntary lapse, and (3) active mortality.

Claim Incidence Assumption

Claim incidence probabilities (i.e., the probability of an insured becoming disabled) were developed using the 2017 Milliman *Long-Term Care Guidelines (Guidelines)* incidence curves, with adjustments for retrospective morbidity improvement using the G2 improvement scale, adverse selection due to past rate increases, and contingent nonforfeiture (CNF) policies. These assumptions were developed based on the following three starting sites of care—assisted living facility (ALF), skilled nursing facility (SNF), or home health care (HHC).

Exhibit A-1 provides a summary of actual-to-expected (A:E) experience by site of care for all products included in the study for claim incidence in policy durations seven and later for each characteristic by which the claim incidence assumption varies. The following items are included:

- Exposure [A] reflects the length of time a covered life is exposed to the risk of becoming disabled (i.e., an exact exposure basis).